# COMMERCIAL ITEM DESCRIPTION

## Test Set, Electrical Power AN/UPM-93( )

The General Services Administration has approved the use of this Commercial Item Description

This Commercial Item Description (CID) delineates the Salient Characteristics (Section A) and the Quality Assurance Requirements (Section B) of commercially available Electrical Power Test Set, modified, as necessary, to meet the requirements specified herein, that is capable of measuring the Voltage and Frequency of Power Circuits

## A SALIENT CHARACTERISTICS

1 Safety Unless otherwise specified herein, the equipment shall comply with the Type II safety requirement specified in MIL-T-28800D

1 1 Personnel Hazards Personnel hazards shall be kept to a minimum The criteria of MIL-T-28800D, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implementec during the course of the contract Verification that compliance with this requirement has been achieved shall be through a visual inspection (see para 1 5 below)

1 2 Edge Rounding Exposed Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radious of 0.04 inch (1 mm) for edges and a minimum of 0 5 inch (12 7 mm) for corners

1 3 Radioactive Materials. Radioactive materials shall not be used (e g , luminous dials/markings, electron tubes, surge arrestors and lenses)

1 4 Weight Any equipment over 2 lbs are considered unacceptable and do not meet the requirement of this specification.

1 5 System Safety Engineering Inspection A visual inspection shall be performed in the presence of a government safety representative to determine that all requirements of 1, 1 1, 1 2, 1 3 and 1 4 have been incorporated in the equipment Inability to meet these requirements shall constitute a failure of the test Results of this inspection shall be included in the Safety Assessment Report (DI-SAFT-80102) if required in the contract.



DISTRIBUTION STATEMENT A. Approved for public release, distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

2 Design and Construction Unless otherwise specified herein, the equipment shall comply with the Type II, Class 5, Style E, Color R, design and construction requirements of MIL-T-28800D All requirements which are automatically invoked by MIL-T-28800D to all types, and specifically to Type II, apply

2 1 Mainframe Plug-in Concept Equipments utilizing externally accessible, removable subassemblies (mainframe plug-in concept) does not meet the requirements of this Commercial Item Description

2 2 Controls All controls required to operate the equipment throughout its specified performance range shall be located on the front panel

3 Parts, Materials, and Processes Unless otherwise specified herein, the equipment shall comply with Type II parts, materials, and processes requirements of MIL-T-28800D

3 1 Restricted Material Equipment shall comply with the restricted material requirements of MIL-T-28800D including the requirement for radioactive deterioration of any material contained in the equipment due to effects of outgassing

3 2 Interchangeability Unless otherwise specified herein, the equipment shall comply with Type II interchangeability requirements of MIL-T-28800D

3 2 1 Marking All replaceable assemblies, subassemblies, modules and printed circuit card assemblies shall be indelibly marked with the Federal Supply Code for Manufacturers (FSCM) and the contractor's part number

4 Marking and Identification Unless otherwise specified herein, the equipment shall comply with Type II marking and identification requirements of MIL-T-28800D

4 . Supplemental Identification Plate The supplemental identification plate specified in MIL-T-28800D shall contain the following data

a Nomenclature
b Contract Number
c Serial Number
d National Stock Number
e US

4.2 Information Plate An information plate affixed to each deliverable equipment shall contain the following data.

a Warranty termination date
b Brief statement of warranty limitations (for example opening a cover voids warranty, calibration adjustments may be made without voiding warranty, defective modules may be returned for warranty service, etc.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

4.3 Battery Warning Indication Equipment containing batteries shall provide a battery warning indication in accordance with MIL-T-28800D, para 3.10.3.1.3.

5 Electrical Power Sources The equipment shall be powered by one 9 volt alkaline battery. The battery shall have a life of at least 200 hours. The battery must be capable of being changed by the operator without removing the case or exposing any internal controls/adjustments. The battery shall be removed from the equipment prior to shipment from the manufacturer and packed in the carrying case.

6 Accessories The following accessories shall be supplied with each equipment:

a. One Government accepted operator's manual.
b. Spare fuses if required for equipment operation (see paras 9.1.10.9.2.4 above).
c. All cables and adaptors necessary for initial turn-on or preoperational check.
d. Other standard accessories, if any, that are offered commercially.
e. A carrying case capable of transporting the meter with all the accessories contained herein.

7 Environmental Requirements Unless otherwise specified herein, the equipment shall comply with the Class 5, Style A, environmental requirements of MIL-T-28800D.

7.1 Vibration The equipment shall comply with Class 5 vibration requirements of MIL-T-28800D, except that the equipment is not required to be operating during the vibration test.

7.2 Electromagnetic Interference (EMI) EMI requirements shall be as specified in MIL-STD-461A, Notice 4(EL) except as noted below. The equipment when tested as specified in para. 7.2.1:

| | | |
|---|---|---|
| CE03 | RE02 | RS03 |
| CE05 | RE02.1 | RS03.1 |

NOTE: The electric field strength and modulation characteristics for fields radiated at unit under test (UUT) shall be as follows:

| Frequency Range (Hz) | E-field | Modulation |
|---|---|---|
| 10 KHz - 2 MHz | 1 | AM, 50%, 1 KHz tone |
| 2 MHz - 30 MHz | 5 | AM, 50%, 1 KHz tone |
| 30 MHz - 76 MHz | 6 | FM, 8 KHz Dev, 1 KHz tone |
| 76 MHz - 400 MHz | 3 | AM, 50%, 1 KHz tone |
| 400 MHz - 2.0 GHz | 3 | Pulse, 0.1 msec, 800 PPS |
| 2.0 GHz - 12.4 GHz | 3 | Pulse, 0.1 msec, 800 PPS |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

7.2 1 EMI Testing. EMI testing shall be in accordance with MIL-STD-462, Notice 3 with exceptions as noted in para. 7 2 above.

8 RAM Requirements

8 1 Reliability requirements The equipment shall comply with the Type II Reliability requirements of MIL-T-28800D (paras 3 13 2 and 4 5 8 1 2) The contractor shall demonstrate his equipment MTBF ($\theta_o$) is equal to or greater than 10,000 hours Reliability tests shall be performed on the first production lot only

8 2 Calibration Calibration interval shall meet or exceed 240 days with 85% of items still within tolerance at the end of the period

8 3 Maintainability Requirements The contractor shall demonstrate his equipment Mean-Time-To-Repair (MTTR) (Ho in MIL-STD-471), is less than or equal to 30 minutes The maximum tolerable MTTR (H1 in MIL-STD-471) shall be 60 minutes The MTTP shall include all the time required to troubleshoot, fault isolate, repair and test the equipment for any malfunction down to the lowest discrete component (resistor, switch, transistor, integrated circuit, nonrepairable assembly, and so forth) of the equipment, but does not include calibration time

9 Performance The following specifications apply to the equipment after a five minute warm-up

9 1 AC Voltage Shall meet the specified performance herein across the full voltage range (see para 9 1 2)

9 1.1 Detection Shall be true RMS detection for signals with crest factors up to 3 1, or greater, across the full voltage range of the equipment

9 1 2 Range Shall be at least 30 millivolts RMS to 300 volts RMS in no less than four ranges

9 1 3 Frequency Response Shall be at least 20 Hz to 5 KHz

9.1 4 Accuracy Shall be at least as specified, or better, over the operating temperature range of 18 to 28 Centigrade, up to 90% humidity

| Accuracy | Frequency |
|---|---|
| 0.5% of input + 2 Counts | 20 Hz to 2 KHz |
| 1% of input + 2 Counts | 2 KHz to 5 KHz |

9 1 5 Volt-Hertz Product Shall be at least 1:10 or better

9 1 6 Resolution Shall be 100 microvolts or less on the lowest range, 1 volt or less on the highest range



Source: https://assist.dla.mil -- Downloaded: 202610-26T09:58Z
Check the source to verify that this is the current version before use.

9 1 7 Response Time Shall be 5 seconds or less to rated accuracy

9 1 8 Input Impedance Shall be 10 megaohms or greater shunted by 100 picofarads capacitance or less

9 1 9 Common Mode Rejection Ratio (CMRR) CMRR shall be at least 60 dB at 50, 60 and 400 Hz and with 1000 ohms unbalance in low lead

9 1 10 Overload Protection Shall provide a minmum of 1000 volts protection (DC or peak AC) or 750 volts RMS

9.2 Frequency Counter Shall meet the specified performance herein across the full frequency range (see para 9 2 1)

9.2 1 Range Shall be at least 40 Hz to 500 Hz.

9.2 2 Accuracy In the temperature range of 18 degrees centigrade to 28 degrees centigrade, up to 90% relative humidity, shall be at least 0 1% or better

9.2 3 Voltage Range Shall be at least 30 millivolts RMS to 300 volts RMS in no less than 4 ranges

9.2 4 Overload Protection Shall withstand at least an input voltage of 750 V RMS or 1000 V DC

9.2 5 Resolution Shall be at least 0 1 Hz

9 2.6 Input Impedance. Shall be at least 10 Megaohms shunted by less than 100 picofarads

9.2.7 Response Time Shall be less than 5 seconds to rated accuracy

9.3 Display The display shall be liquid crystal with at least three and a half digits, electronic digital display It shall display voltage in units of volts and millivolts RMS and frequency in units of hertz.

9.4 Input Connector The input shall be a recessed banana male or female connector There shall be an input terminal for volts and frequency measurement and a common terminal

9 5 Ranging The equipment in all modes of operation shall be manual ranging

9 6 Temperature Coefficient For temperatures outside the 18 degrees centigrade to 28 degrees centigrade range, less than or equal to 0.1% of specified accuracy (see paras 10.1.4 and 10.2.2) shall be cumulatively added for every degree below 18 degrees centigrade and above 28 degrees centigrade



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

9.7 Other Capabilities. The other performance capabilities of the equipment that are standard features of the commercial equipment, however, not covered by the above paragraphs of this CID

10 Preservation, Packaging, Packing, labeling and marking The preservation, packaging, packing, labeling and marking (other than those required in paragraph 4 above) shall be as specified in the contract or order

5 QUALITY ASSURANCE PROVISIONS

1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

2 Preconditioning Requirement (Burn-in) Each deliverable equipment shall be subjected to one of the following preconditioning requirements

a A minimum 48 hours on-time burn-in period prior to Acceptance Testing During the last 24 hours of on-time burn-in, the equipment shall operate failure free up until this time, the equipment will be allowed to accumulate failures Each equipment which fails during the final 24-hour period shall be repaired and returned to test until it successfully survives a 24-hour period without failure Failures which occur during the burn-in test shall be noted and reported, but shall not count toward the establishment of equipment MTBF Prior to burn-in the Satisfactory Operation Test (SOT) shall be conducted Daily Satisfactory Operation Checks (SOC) shall be conducted For the last 24-hour failure-free period, a complete SOT shall be conducted prior to and after the period The SOC and SOT are defined by MIL-T-28800D, paras 6 3 2 24 and 6.3 2 25 While under burn-in the equipment shall be subjected to the following conditions.

Ambient Temperature 40°C +5°C -0°C

Input Voltage Nominal specified voltage +5% -2%

Equipment On-Off Cycling Equipment on for 8 hours and off for 4 hours This cycle shall continue until a minimum 48 hours on-time is accumulated, with the last 24 hours of on-time, failure free



Source: https://assist.dla.mil -- Downloaded: 202640-26T09:58Z
Check the source to verify that this is the currentversion before use.

b Stress Screen A minmum 15 minute vibration period followed by a minimum 10 thermal cycle period, shall constitute stress screen, prior to acceptance testing For the vibration period and the last three thermal cycles of stress screen, each equipment must survive failure free Each equipment which fails during the vibration period or the last three thermal cycles shall be repaired and returned to test until it successfully survives these conditions Failures which occur during the combined vibration/thermal stress screen shall be noted and reported but shall not count toward the establishment of equipment MTBF An SOT shall be conducted prior to vibration and thermal cycling An SOT shall also be conducted during a 15 minute ambient temperature (25 C) test period, following each of the last three thermal cycles While under stress screen, the equipment shall be subjected to the following conditions

Number of Thermal Cycles 10

Temperature Cycle -40°C to +75°C

Temperature Rate of Change 5°C/Min

Input Voltage Nominal Specified Voltage +5%, -2%

Equipment On-Off Cycling Equipment on at 0°C and off at 50°C Equipment shall not operate during chamber cool down

Vibration, Sinusoidal, linear sweep Sweep times shall be for a low to high to low frequency variation for each increment Equipment shall not operate during vibration

| Frequency Increment Hz (High-Low) | Vibration Level (Inch Pk-Pk) | Vibration Time per Increment (Min ) |
|---|---|---|
| 5-15 | 0 06 | 7 |
| 15-25 | 0 04 | 3 |
| 25-55 | 0 02 | 5 |

Vibration Axis Single axis, perpendicular to the plane of normal bench usage

3 Quality Conformance Inspection Unless otherwise required herein the test methods, procedures, and conditions used for quality conformance inspection shall be as as specified in MIL-T-28800D Equipment shall be subjected to preconditioning requirement, para 2 above, prior to production inspection



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

4 Production Inspection Each equipment shall be subjected to the Group A inspection of Table 1 to ensure qualitatively the proper functioning of the equipment, including all operating controls and conformance with safety requirements Group A testing (Table 1) is a reduced amount of testing Its purpose is to ensure that all functions and modes of operation of the equipment are evaluated without extensively checking each parameter as required in the Group B test (Table II)

4 1 Production Control Tests The Group B tests of Table II shall be performed on equipments that have passed Group A inspection, para 4 above

4 1.1 Production Control Sampling Plan From each lot a random sample of equipment shall be selected and subjected to the Group B tests Sampling shall be in accordance with MIL-STD-105D, special inspection level S-3, AQL 1 5%

4 2 Environmental Tests The Group C tests of Table III shall be performed on equipments which have passed the required tests in accordance with para 4.1 above

4 2 1 Environmental Sampling Plan From the first production lot only a random sample of equipment shall be selected and subjected to the Group C tests The sampling shall be in accordance with MIL-STD-105D, special inspection S-1, AQL 4 0%.

4 3 Reliability Tests The Group D tests of Table IV shall be performed on equipments which have passed the required tests in accordance with para 4 1 above

4 3 1 Reliability Tests Sampling Plan From the first production lot only a random sample of 10 equipments shall be selected and subjected to Group D tests

4 4 Other Tests The Group E design tests of Table V shall be performed on equipments which have passed the required tests in accordance with para 4 1 above

4.4 1 Group E Design Tests Sampling Plan From the first production lot only, 1 equipment shall be randomly selected and subjected to the Group E tests with the exception of maintainability testing which requires a dedicated second sample

5 Reliability The reliability tests shall be performed as follows

5 1 Test Level While under test, the equipment shall be subjected to conditions specified in para 4 5 8 1 2 of MIL-T-28800D (Test Plan IIID of MIL-STD-781D is invoked).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

5.2 Quality Conformance Inspection The Group D reliability test shall be performed on the first production lot only From those units of the first production lot that have passed the required Group A and B tests, a random sample of ten (10) units shall be selected for Group D testing

5 3 Test Length Testing shall continue until the total unit hours together with the total count of relevent equipment failures permit either an accept or reject decision Only equipment 'ON' time may be used in MTBF or longevity determinations Testing shall be monitored in such a manner that the time to failure may be estimated with reasonable accuracy No single equipment 'ON' time shall be less than one-half the average operating time of all equipments 'ON' time

5 4 Reliability Examination and Test Method The following inspections shall be used to verify equipment operation during reliability testing The SOT of para 2 above shall be used to confirm proper equipment operation prior to and following reliability testing The SOT shall also be performed weekly to verify equipment operation The SOC of para 2 above shall be used to monitor proper operation of the equipment daily

6 Maintainability Maintainability testing shall be conducted in accordance with para 4 5 7.1 1 of MIL-T-28800D

TABLE I

GROUP A PERFORMANCE TESTS

| REQUIREMENT | PARAGRAPH (Section A) |
|---|---|
| Safety | 1 |
| Personnel Hazards | 1 1 |
| Edge Rounding Exposed | 1.2 |
| Marking | 3 2 1 |
| Marking and Identification | 4 |
| Supplemental Identification Plate | 4 1 |
| Information Plate | 4 2 |
| Battery Warning Indication | 4 3 |
| Electrical Power Sources | 5 |
| Accessories | 6 |
| Range | 9 1 2 |
| Range | 9.2 1 |
| Display | 9.3 |
| Input Connector | 9 4 |
| Ranging | 9 5 |
| Other Capabilities (Selected Features) | 9 7 |



Source: https://assist.dla.mil -- Downloaded: 202610-26T09:58Z
Check the source to verify that this is the current version before use.

TABLE II
GROUP B PERFORMANCE TESTS

| REQUIREMENT | PARAGRAPH (Section A) |
|---|---|
| Detection | 9 1 1 |
| Frequency Response | 9 1 3 |
| Accuracy | 9 1 4 |
| Volt-Hertz Product | 9 1 5 |
| Resolution | 9 1 6 |
| Response Time | 9 1 7 |
| Input Impedance | 9 1 8 |
| CMRR | 9 1.9 |
| Overload Protection | 9 1 10 |
| Accuracy | 9 2 2 |
| Voltage Range | 9 2 3 |
| Overload Protection | 9 2 4 |
| Resolution | 9 2 5 |
| Input Impedance | 9 2 6 |
| Response Time | 9 2 7 |
| Temperature Coefficient | 9 6 |
| Other Capabilities (not covered under Group A) | 9 7 |

TABLE III

GROUP C ENVIRONMENTAL TESTS

Environmental Tests — MIL-T-28800D and Paragraph 7, Section A, of this Description

| TEST | REQUIREMENT |
|---|---|
| Temperature, Non-operating | MIL-T-28800D |
| Temperature, Operating | MIL-T-28800D |
| Humidity | MIL-T-28800D |
| Altitude, Non-Operating | MIL-T-28800D |
| Vibration | MIL-T-28800D, see para 7 1 for exception |
| Pulse Shock | MIL-T-28800D |
| Bench Handling | MIL-T-28800D |
| *Fungus Resistance | MIL-T-28800D |
| Sand and Dust Resistance | MIL-T-28800D |
| Splashproof | MIL-T-28800D |
| Transit Drop | MIL-T-28800D |
| Salt Exposure, Structural Parts | MIL-T-28800D |
| Salt Exposure, Enclosure | MIL-T-28800D |

* Verification testing not required, but certificate of conformance is required.



Source: https://assist.dla.mil -- Downloaded: 202610-26T09:58Z
Check the source to verify that this is the current version before use.

TABLE IV

GPOUP D RELIABILITY TESTS

| TEST | PARAGRAPH (Section A) |
|---|---|
| Reliability | 8 1* |

* Also para 5, section B

TABLE V

GROUP E DESIGN TESTS

| REQUIREMENT | PARAGRAPH (Section A) |
|---|---|
| Radioactive Materials | 1 3 |
| Weight | 1 4 |
| System Safety Engineering Inspection | 1 5 |
| Design and Construction | 2 |
| Mainframe Plug-in Concept | 2 1 |
| Controls | 2 2 |
| Parts, Materials and Processes | 3 |
| Restricted Material | 3 1 |
| Interchangeability | 3 2 |
| EMI | 7 2 |
| Maintainability | 8 3 |
| | PARAGRAPH (MIL-T-28800D) |
| Accessible Potentials | 3 9 3 1 |
| Ventilation opening | 3 9 3 3 |
| Equipment Opening | 3 9 3 4 |
| Thermal Hazard Protection | 3 9 3 5 |
| Warning Markings | 3 9 3 6 |
| Guards and Shields | 3 9 3 7 |
| Toxic Material | 3 9.3 9 |
| Compatibility of Material | 3 9 3 10 1 |

MILITARY INTERESTS.

Custodian:
Army - CR

CIVIL AGENCY COORDINATION ACTIVITY

GSA-FSS

PREPARING ACTIVITY:
Army - CR

(Project 6625-A961)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.